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SCHOOL OF CIVIL, ENVIRONMENTAL, AND LAND MANAGEMENT
ENGINEERING



MASTER OF SCIENCE IN CIVIL ENGINEERING

PROJECT III: SEISMIC VULNERABILITY OF AN EXISTING
MASONRY BUILDING

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CHAPTER 1

OVERVIEW

1.1 INTRODUCTION:

The current project is the solution to the third assignment of the master's degree in Civil Engineering course Earthquake Resistant Design at Politecnico di Milano. The primary purpose is to analyze the seismic risk of brick construction. The demise of buildings is a major source of casualties during an earthquake. Nowadays, building rules include specifications to avoid harm from the earthquakes of mild and medium strength, and to avoid collapse (allowing severe damage though) due to high-intensity earthquakes. If the structure possesses ductile properties, huge deformations may occur, but not to the point of collapse, which is the goal of seismic prescriptions. The purpose of the codes is to ensure ductile behavior. It is obvious that in the past, the level of concern regarding seismic consequences on structures was substantially lower. This feature, together with the usage of materials that result in brittle behavior, is the cause of the prevalence of brittle structures that are not safe from the effects of an earthquake. As a result, Italian creatures have evolved strategies for assessing building susceptibility.

Risk analysis from seismic activities is critical for existing structures in nations where existing buildings are extensively distributed. Verifications for existing structures built of masonry or other materials are frequently necessary for design practice. Some nations, such as Italy, have many masonry structures that are historically and culturally significant. Given that most historic buildings have little knowledge and many uncertainties, multiple types of verification procedures given by the Eurocode and National Codes can be used to ensure the safety of structures. A seismic analysis determines if a building's seismic capacity exceeds its seismic demand. Specifically, linear analyses are usually performed in terms of forces, i.e., it is checked that each structural element is not subjected to a force greater than its strength capacity, whereas nonlinear analyses are performed by considering deformation limits in each structural element, and the verification is done at the global level, i.e., it is checked that a building's maximum nonlinear displacement demand is less than its displacement capacity.

It was therefore necessary to set criteria for masonry structures considering the various knowledge levels in the new Italian structural regulations (currently published in the NTC, 2008, document and in the appropriate recommendations Circ. NTC08). The goal of this project is to do a seismic evaluation of an existing brick structure using a simplified global way of analysis – The level 1 approach. The idea behind this method is to compute the return duration that corresponds to the requirements of attaining no collapse condition and the ground acceleration that corresponds to structural resistance, which can then compare with the earthquake's reference value. The computation of the safety index gives information about the safety of the structure, which is given as the ratio between the structural capacity and the action. To consider a structure safe, the safety index should be at least or greater than the unit value. This safety index is calculated considering some assumptions and simplifications for the considered structure.

Italian National building Code and seismic design prescriptions:

According to the Italian National Code, NTC 2018 and its previous version 2008, to chapter 8 dedicated to seismic behavior and design, an existing building is the one with the structure already built and usable now of a safety check, survey, and similar. The safety check and the

consequent design for intervention must consider key features such as the history of the fabrique and the level of knowledge, the presence of defects not visible to the human eye, the consequences (also not already appeared) of external forces, modifications, and decay with respect to the original structure. So, depending on the availability of information, it is important to know about an existing building:

- Geometry and constructive details
- Mechanical properties of materials and soil
- Acting loads on the structure.

Safety Evaluation:

According to NTC 2018, chapter 8.3, safety evaluation is a quantitative technique capable of defining the degree of forces that the evaluated structure can endure, guaranteeing the minimum level of safety required by the codes. If this minimum level is not met, structural safety is given via adjusting the structure's global behavior as well as through local actions. These examinations are especially difficult in the case of cultural heritage structures due to the constraints imposed by the desire to preserve the structure's integrity and the history concealed therein.

For these reasons, checks must not be invasive and must be performed following a thorough job of information acquisition and a series of non-intrusive examinations to identify the structural behavior and circumstances of the current structure. The Italian National Code requires inspections for SLU (Ultimate Limit State, Stato Limite Ultimo) and SLE (Serviceability Limit State, Stato Limite di Esercizio), with all loads operating on the structure presumed to be equal in new and existing structures. These checks attempt to define the structural behavior of existing structures and, for the purposes of this research, seismic activity, and subsequent adjustments to assure occupant safety. After a first step of checking the condition of the structure also through structural models, according to chapter 8.4 of the NTC 2018, there are different interventions to be designed against seismic events for an existing structure. The categories for the intervention are:

- Local intervention or restoration (intervento di riparazione o locale)
- Seismic improvement intervention (intervento di miglioramento)
- Retrofitting intervention (intervento di adeguamento).

Definition of Model Analysis:

According to NTC 2018, there are specified measures to take to generate a model that is as near to the true condition of the existing building as feasible. This approach given by the national code is quite like and serves as the foundation for what the Cultural Heritage Ministry proposes and will describe in the next chapters. Furthermore, even though this model is dealing with existing structures, the loads to be applied for the inspections are the ones specified by the code for new builds.

Primarily, according to NTC, it is critical to have trustworthy information on the construction history of the researched building, based on archives or on-site studies, to comprehend the process of evolution of the structure and all the acts that might impact its behavior. A geometric relief is required to determine the resistant structure, its circumstances, and material quality. The mechanical characteristics of the materials can be evaluated using on-site testing

(diagnostic tests and visual inspection) and laboratory tests; the tests must be recorded and confined to the essential ones.

Following the gathering of information, three levels of knowledge (livelli di conoscenza) can be defined according to the Italian National Code: LC1, LC2, and LC3. These stages are determined by the quality of information on the geometry, structural details, connection, and mechanical characteristics of materials and elements derived from the preceding knowledge stream.

1.2 SITE DESCRIPTION:

Some information is conducted with Italian national annex NTC-2008. This building is in the city of Udine. Udine is a city and commune in north-eastern Italy, in the middle of the Friuli-Venezia Giulia region, between the Adriatic Sea and the Alps. Udine is 113m above sea level. Given soil type C. Soil C type constituted by deposits of coarse-grained soils, medium dense, or fined grained soils. According to website data, all the towns of the city fall under the category of seismic zone II. Environmental conditions have been considered “aggressive,” so it has been used in the exposure class XC1. Our site is 113m above sea level.

GEOMETRY:

The chosen structure for analysis is a two-story masonry building with a wooden roof for residential usage. As previously stated, the structure is in Udine. The structure has a non-symmetric plan with respect to the reference axes (X, Y) and a non-symmetric wall distribution. This structure's structural system is distinguished by an ordinary masonry wall solution consisting of eleven walls in the longitudinal (X) direction and seven walls in the transversal (Y) direction. The structure's plan view is depicted below.

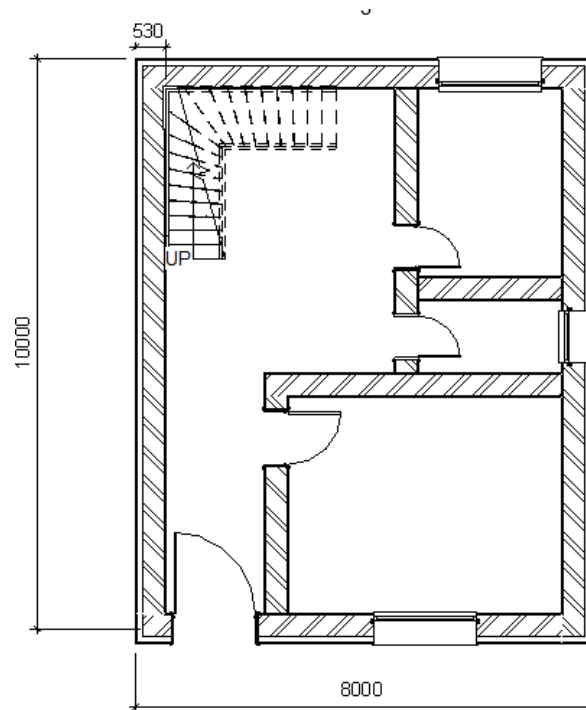


Figure 1: Ground Floor Plan

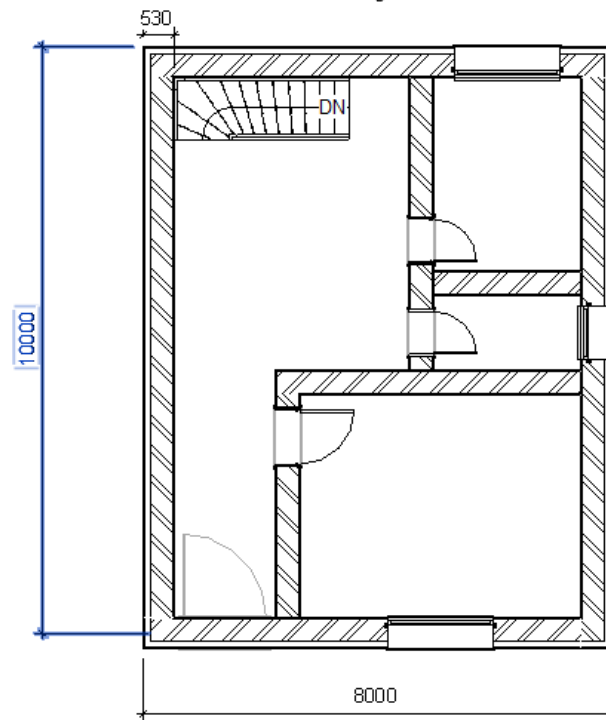


Figure 2: First Floor Plan

As mentioned above, the building has a regularity in elevation. In the pictures below are shown the front and lateral view which represents the characteristic cross-section for two principal directions of the buildings, in x and y direction.

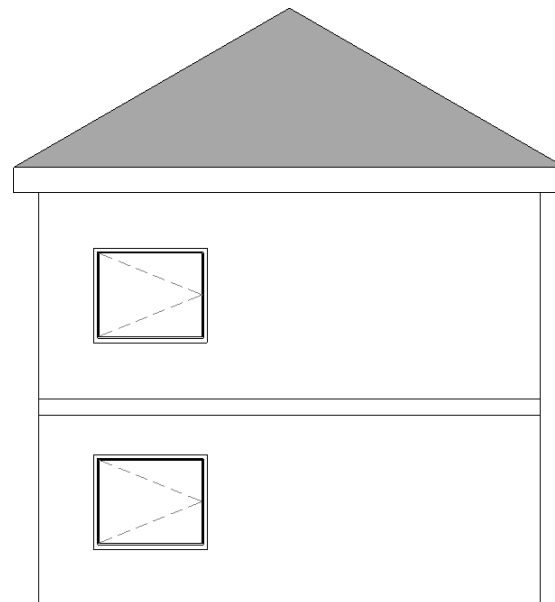


Figure 3: Elevation

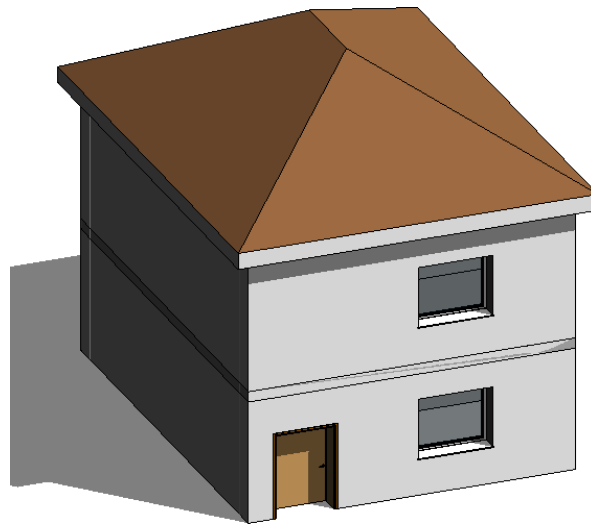


Figure 4: 3D Plan

1.3 MATERIAL:

Masonry is the construction of structures out of discrete pieces that are commonly set in and joined together with mortar; the term masonry can also refer to the components themselves. Brick, building stone such as marble, granite, limestone, cast stone, concrete block, glass block, and adobe are typical masonry construction materials. Masonry is a very long-lasting type of building. However, the materials used, the quality of the mortar and workmanship, and the pattern in which the units are joined can all have a significant impact on the overall durability of the masonry building. A mason or bricklayer is someone who builds masonry. Both are considered construction trades. The enormous variety of conceivable permutations formed by the geometry, nature, and placement of units, as well as the features of the joints, casts doubt on the word masonry's accuracy in identifying a single structural material. Masonry strength is determined by the unit and mortar qualities, as well as the building procedure. The compressive strength of stone units, for example, can range from 5 MPa (poor quality limestone) and even less for tuff to over 130 MPa (high-quality limestone) and much more for granite or marble.

The strength of the mortar varies greatly and is determined by the proportions of its components (cement, lime, sand, soil, and water) used in the mix. The compressive strength of existing masonry mortar. Even if weaker and stronger mortars may be found, structures can be 1.5-3.5 MPa. Furthermore, the loading impacts the strength and fracture modes of brickwork. Load direction and combination. Nonetheless, the mechanical behavior of various. Some characteristics of masonry are as follows: high specific mass, low tensile strength, and low to medium density. Shear strength is moderate, while ductility is poor. Stone masonry-specific mass can range from between 1700 and 2200 kg/m³.

KNOWLEDGE AND CONFIDENCE FACTOR:

The mechanical reference values, maximum and lowest, and average specific weight for each masonry type may be obtained from Cr. NTC 2008 number 617 dated 02/02/2009 (Tab. C8A.2.1).

Considering equal to one the level of confidence for the verification procedure the values for resistance are taken as the minimum ones from the tables and the elasticity modulus is taken

as the mean value (consideration for the existing buildings). For the considered wall, made of semi-solid brickwork (“Muratura in mattoni semipieni - Doppio UNI”), the properties are defined as follows:

Due to the restricted knowledge level in each inquiry category and several assumptions, a safety value $F_c = 1.35$, which has the unit knowledge level, has been assumed. Another decrease was implemented owing to the safety coefficient $M = 0.8$ from NTC2008 table C8A.2.2 for brickwork with a low-quality core. We took all values from the figure below.

- $f_m = 5.0 \text{ MPa}$
- $\tau_0 = 0.24 \text{ MPa}$
- $E = 3500 \text{ MPa}$
- $G = 875 \text{ MPa}$

Tipologia di muratura	f_m (N/cm ²)	τ_0 (N/cm ²)	E (N/mm ²)	G (N/mm ²)	w (kN/m ³)
	Min-max	min-max	min-max	min-max	
Muratura in pietrame disordinata (ciottoli, pietre erratiche e irregolari)	100 180	2,0 3,2	690 1050	230 350	19
Muratura a conci sbazzati, con paramento di limitato spessore e nucleo interno	200 300	3,5 5,1	1020 1440	340 480	20
Muratura in pietre a spacco con buona tessitura	260 380	5,6 7,4	1500 1980	500 660	21
Muratura a conci di pietra tenera (tufo, calcarenite, ecc.)	140 240	2,8 4,2	900 1260	300 420	16
Muratura a blocchi lapidei squadriati	600 800	9,0 12,0	2400 3200	780 940	22
Muratura in mattoni pieni e malta di calce	240 400	6,0 9,2	1200 1800	400 600	18
Muratura in mattoni semipieni con malta cementizia (es.: doppio UNI foratura $\leq 40\%$)	500 800	24 32	3500 5600	875 1400	15
Muratura in blocchi laterizi semipieni (perc. foratura $< 45\%$)	400 600	30,0 40,0	3600 5400	1080 1620	12
Muratura in blocchi laterizi semipieni, con giunti verticali a secco (perc. foratura $< 45\%$)	300 400	10,0 13,0	2700 3600	810 1080	11
Muratura in blocchi di calcestruzzo o argilla espansa (perc. foratura tra 45% e 65%)	150 200	9,5 12,5	1200 1600	300 400	12
Muratura in blocchi di calcestruzzo semipieni (foratura $< 45\%$)	300 440	18,0 24,0	2400 3520	600 880	14

Table 1: Masonry typologies and Characteristics Value

CHAPTER 2

LOAD ANALYSIS:

2.1 LOADS ON ELEMENTS:

Loads that act on the structure are:

1. Permanent load (G_i):

- Self-weight of the masonry (structural load)
- Floors, roofs, stairs, finishes (non-structural load)

2. Live load (Q_v):

- Snow, wind etc.

ROOF SLAB:

Slab of the roof is characterized by the mass per unit area of the value $297.26 \text{ (kg/m}^2\text{)}$, taken into design consideration as an equivalent distributed load with a value of G_R , slab = $2.970 \text{ (kN/m}^2\text{)}$. The variable load in this case is q_R , slab = $0.5 \text{ (kN/m}^2\text{)}$. The section of roof slab is shown in the picture below:

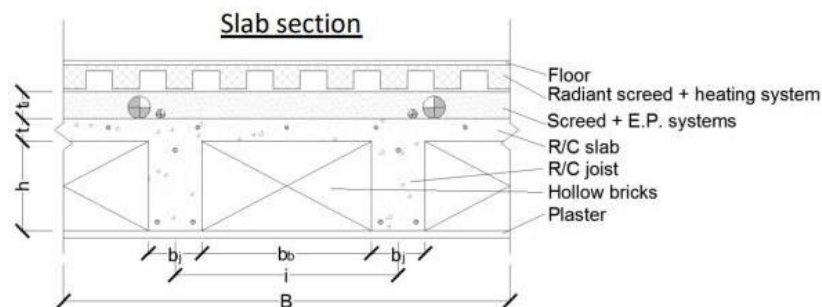


Figure 5: Roof Slab

ROOF:

The roof structure is characterized by the mass per unit area of a value $458 \text{ (kg/m}^2\text{)}$ in the horizontal plane. The roof is lying in the inclination of 30 %. Considering the inclination of the roof the projected load value is equal to $G_{\text{roof}} = 4.8 \text{ (kN/m}^2\text{)}$ and the live loads of a value $q_{\text{roof}} = 0.5 \text{ kN/m}^2$. In the figure below is presented the cross section of the roof.

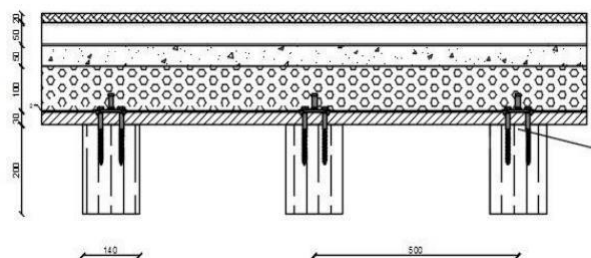


Figure 6: Roof

FLOOR SLAB:

Floor slab is characterized by the mass per unit area of the value of $295.5 \text{ (kg/m}^2\text{)}$, taken into consideration as an equivalent distributed load value of $G_{\text{slab}} = 2.950 \text{ (kN/m}^2\text{)}$. The variable loads in this case are considered to be $q_{\text{slab}} = 2.00 \text{ (kN/m}^2\text{)}$. The cross section of the floor slab is given in figure below:

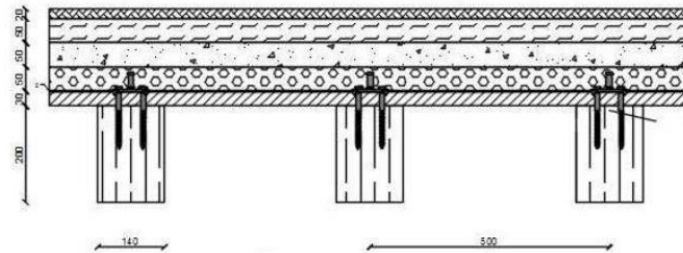


Figure 7: Floor Slab

STAIRS:

The stairs used for the connection of the two floors of the building are made of reinforced concrete by using an anchorage to the lateral walls of the structure. Considering the inclination of the stairs of the value 28° , the projected unit area will be under the projected mass of the value of $900 \text{ (kg/m}^2\text{)}$ taken into consideration as an equivalent distributed load of a value of $G_{\text{stairs}} = 9.0 \text{ (kN/m}^2\text{)}$. For these considered stairs the live loads are supposed to be equal to $q_{\text{stairs}} = 4.0 \text{ (kN/m}^2\text{)}$. The cross section of the stairs is given in the following figure:

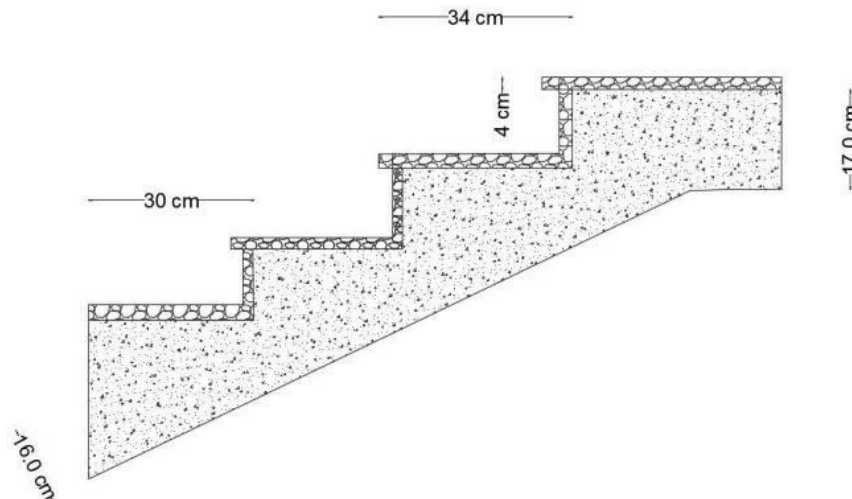


Figure 8: Stairs

MASONRY WALLS:

Considered structure is characterized by the two kinds of masonry walls: the ones used for the internal and external side. Difference between these two types of walls is in their thickness, considering the external walls to have an additional layer of insulating material of

10 cm, having thermo-acoustic properties. For the design these masses are taken as equivalent distributed load of the value of $G_{\text{wall,ext}} = 6.42 \text{ (KN/m}^2 \text{)}$ and $G_{\text{wall,int}} = 6.38 \text{ (KN/m}^2 \text{)}$. The schemes for the two walls cross sections are given in figures below:

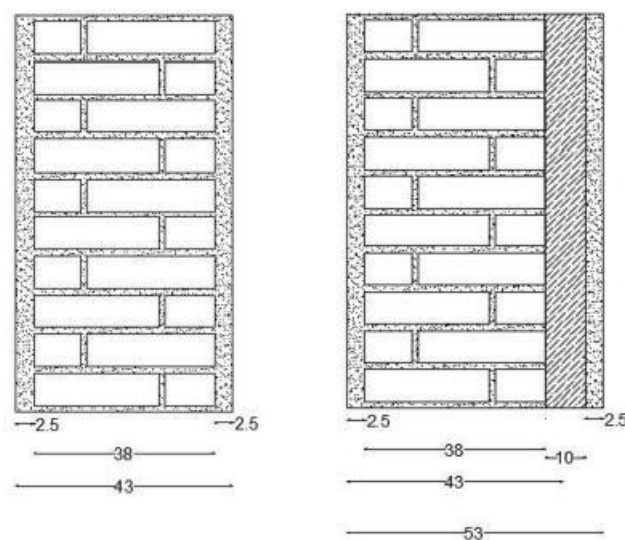


Figure 9: Cross section of internal and external masonry walls

EXTERNAL WALL			
Element	t(m ²)	spec. density	p [kg/m ²]
Double Brick	0.38	1450	551
Internal Coat	0.025	2000	50
External Coat	0.025	2000	50
Insulation	0.1	35	3.5
			6.42 kN/m²

Table 2: External masonry wall elements and their properties

INTERNAL WALL			
Element	t(m ²)	spec. density	p [kg/m ²]
Double Brick	0.38	1450	551
Internal Coat	0.025	2000	50
External Coat	0.025	2000	50
			6.38 kN/m²

Table 3: Internal masonry wall elements and their properties

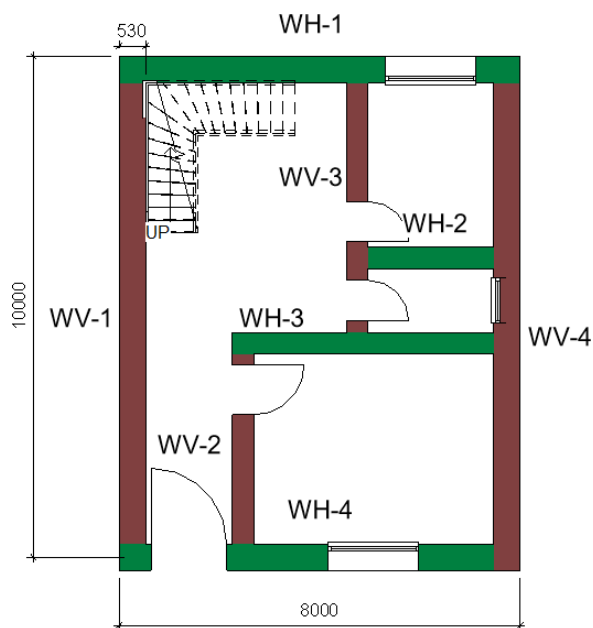


Figure 10: External and Internal Walls

Horizontal walls								
Wall	Thickness (m)	Length (m)	A (m ²)	hF ₀ (m)	hF ₁ (m)	P (kN/m ²)	SW F ₀ (KN)	SW F ₁ (KN)
WH1	0.53	7	3.71	3.3	3.3	6.42	23.8182	23.8182
WH2	0.43	2.5	1.075	3.3	3.3	6.38	6.8585	6.8585
WH3	0.43	5.215	2.2424	3.3	3.3	6.38	14.30683	14.30683
WH4	0.53	6	3.18	3.3	3.3	6.42	20.4156	20.4156

Table 4: Horizontal Walls

Vertical walls								
Wall	Thickness (m)	Length (m)	A (m ²)	H F ₀ (m)	h F ₁ (m)	P (kN/m ²)	SW F ₀ (KN)	SW F ₁ (KN)
WV1	0.53	10	5.3	3.3	3.3	6.42	34.026	34.026
WV2	0.43	3.265	1.403	3.3	3.3	6.38	8.95720	8.95720
WV3	0.43	4.45	1.9135	3.3	3.3	6.38	12.2081	12.2081
WV4	0.53	10	5.3	3.3	3.3	6.42	34.026	34.026

Table 5: Vertical Walls

SNOW LOAD:

According to “Eurocode 1 - Actions on structures - Part 1-3: General actions - Snow loads”, the total snow load acting on the roof is evaluated with the following expression

$$S = S_k \cdot \mu_i \cdot C_e \cdot C_t$$

Where,

- S is the snow load on the roof
- μ_i is the snow load shape coefficient.

- S_k is the characteristic value of the snow load on the ground reference for a period of return of 50 years
- C_e is the exposure coefficient
- C_t is the thermal coefficient.

For the snow load, it has been considered that our area lies in zone 1-A of the Alpine region at 113 m above the sea level, lower than (200 m) as shown in the figure below.

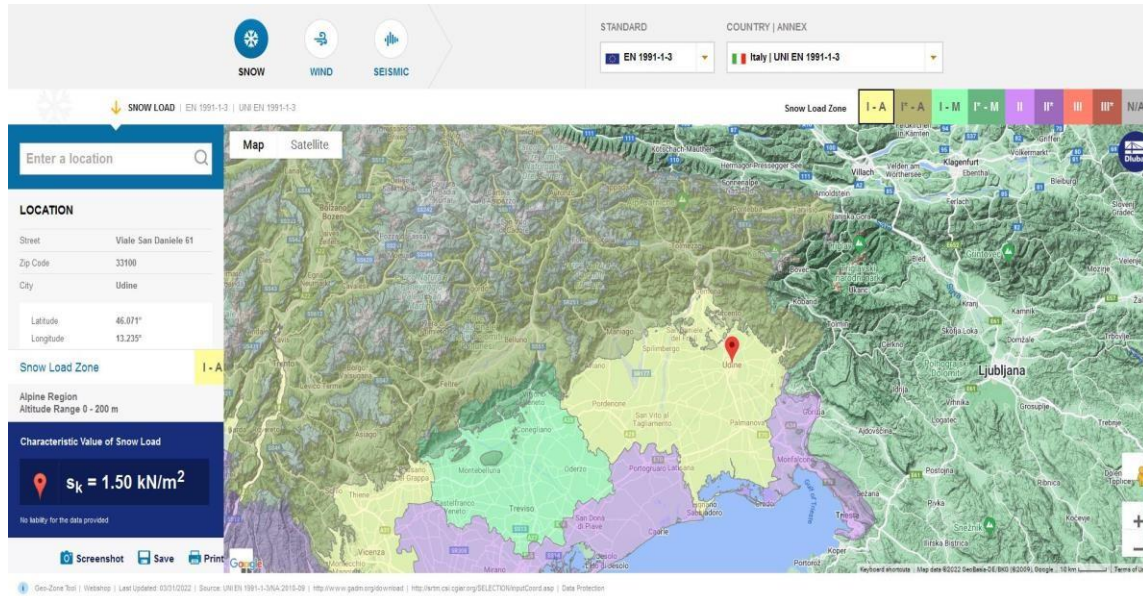


Figure 11-Udine region Snow load characteristics

Conventionally, the effects due to the action of the snow are followed by S symbol and then, the resulting load per unit area is defined by the symbol S_k .

CHARACTERISITICS VALUE OF SNOW LOAD:

The Italian snow map is divided into four different homogeneous climatic zone; in each climatic region we have a given load-altitude correlation formula, and this is given in the following table.

Zone	Values	ASL
Zone I Alpine	$S_k = 1.5 \text{ kN/m}^2$ $S_k = 1.39 [1 + (as/728)^2] \text{ kN/m}^2$	$as \leq 200\text{m}$ $as > 200\text{m}$
Zone II Mediterranean	$S_k = 1.5 \text{ kN/m}^2$ $S_k = 1.39 [1 + (as/602)^2] \text{ kN/m}^2$	$as \leq 200\text{m}$ $as > 200\text{m}$
Zone III	$S_k = 1.0 \text{ kN/m}^2$ $S_k = 1.39 [1 + (as/481)^2] \text{ kN/m}^2$	$as \leq 200\text{m}$ $as > 200\text{m}$
Zone IV	$S_k = 0.6 \text{ kN/m}^2$ $S_k = 1.39 [1 + (as/481)^2] \text{ kN/m}^2$	$as \leq 200\text{m}$ $as > 200\text{m}$

Table 6-Italian Characteristics Snow Load Zones

As our building area is in Zone 1 and it is 113m above sea level, so our characteristics snow load coefficient is S_k is 1.5 kN/m^2 . As we can see from above picture taken from a website.

EXPOSURE COEFFICIENT:

The exposure coefficient C_e considers the place in which the structure is located: dealing with a “normal” zone, the Code Table 5.1 suggests C_e equals to 1. The exposure coefficient C_e should be used for determining the snow load on the roof. The choice for C_e should consider the future development around the site and it should be taken as 1.0 unless otherwise specified for different topographies.

Topography	C_e
Windswept ^a	0,8
Normal ^b	1,0
Sheltered ^c	1,2

^a Windswept topography: flat unobstructed areas exposed on all sides without, or little shelter afforded by terrain, higher construction works or trees.

^b Normal topography: areas where there is no significant removal of snow by wind on construction work, because of terrain, other construction works or trees.

^c Sheltered topography: areas in which the construction work being considered is considerably lower than the surrounding terrain or surrounded by high trees and/or surrounded by higher construction works.

Figure 12- Exposure Coefficient from NTC-2018

SNOW LOAD SHAPE COEFFICIENT:

Since there are not specific indications regarding roof's geometry, it has been used the worst-case scenario (Table 5.2) of roof pitch inclination on the horizontal plane ($0^\circ \leq \alpha \leq 30^\circ$), which sets μ_i equal to 0.8.

Angle of pitch of roof α	$0^\circ \leq \alpha \leq 30^\circ$	$30^\circ < \alpha < 60^\circ$	$\alpha \geq 60^\circ$
μ_1	0,8	$0,8(60 - \alpha)/30$	0,0
μ_2	$0,8 + 0,8 \alpha/30$	1,6	--

Table 7- Snow Load Shape Coefficient

THERMAL COEFFICIENT:

The thermal coefficient C_t should be used to account for the reduction of snow loads on roofs with high thermal transmittance ($> 1 \text{ W/m}^2\text{K}$), but in our case we can assume a value of $C_t = 1$ as temperature in the case of snow is typically low so we are assuming 1. So, our total snow load comes out to be,

$$S = S_k \cdot \mu_i \cdot C_e \cdot C_t$$

$$S = 1.5 \cdot 0.8 \cdot 1 \cdot 1 = 1.2 \text{ kN/m}^2$$

SNOW LOAD	
a_s [m]	113
q_{sk} [kN/m ²]	1.50
C_e	1
C_t	1
μ	0.8
q_s [kN/m ²]	1.20

Table 8: Snow Load

WIND LOAD:

Wind direction is considered to be predominantly horizontal. The load is computed according to Eurocode -1(2005) in both x and y directions, Wind load will be calculated with this formula.

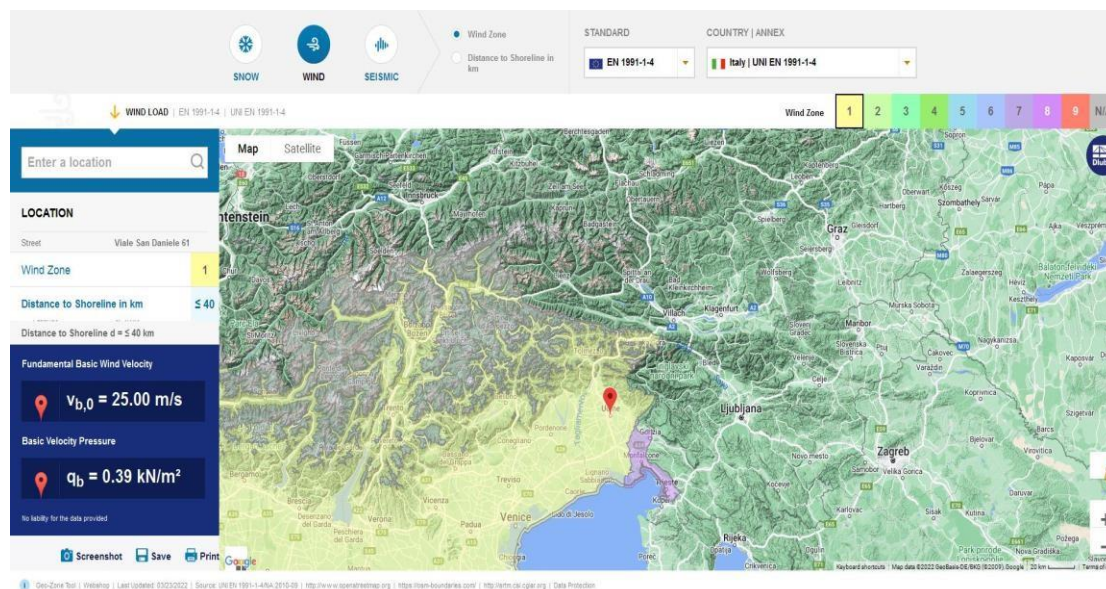


Figure 13- Wind Zoning of Udine Region

Udine is in wind zone 1 and distance to shoreline is less than 40km.

$$p = q_b \cdot C_e \cdot C_p \cdot C_d$$

Where,

- q_b is the peak velocity pressure
- C_e is the exposure coefficient
- C_p is aerodynamic co- efficient depending on the geometry and orientation of the building. This value is obtained by wind tunnel test
- C_d is the dynamic coefficient which takes account of the reductive effects associated with non-contemporaneity of maximum local pressures and amplification effects due to structural vibration

BASIC WIND VECLOCITY:

The Basic wind velocity, V_b , is the characteristic value of the wind speed at 10 m above the ground on a plot of category II exposure, averaged over ten minutes and referred to a period of return of 50 years.

According to NTC-18 we can use this table to verify the basic wind velocity given by website as attached the screenshot above and by this formula we can find V_b .

$$V_b = V_{b,0} \quad a_s \leq a_0$$

$$V_b = V_{b,0} + k a (a_s - a_0) \quad a_0 < a_s \leq 1500m$$

Zona	Descrizione	$v_{b,0}$ [m/s]	a_0 [m]	k_s [1/s]
1	Valle d'Aosta, Piemonte, Lombardia, Trentino Alto Adige, Veneto, Friuli Venezia Giulia (con l'eccezione della provincia di Trieste)	25	1000	0.010
2	Emilia Romagna	25	750	0.015
3	Toscana, Marche, Umbria, Lazio, Abruzzo, Molise, Puglia, Campania, Basilicata, Calabria (esclusa la provincia di Reggio Calabria)	27	500	0.020
4	Sicilia e provincia di Reggio Calabria	28	500	0.020
5	Sardegna (zona a oriente della retta congiungente Capo Teulada con l'Isola di Maddalena)	28	750	0.015
6	Sardegna (zona a occidente della retta congiungente Capo Teulada con l'Isola di Maddalena)	28	500	0.020
7	Liguria	28	1000	0.015
8	Provincia di Trieste	30	1500	0.010
9	Isole (con l'eccezione di Sicilia e Sardegna) e mare aperto	31	500	0.020

Table 9- Basic Wind Velocities with Italian regions

As we are in Udine city, so our region is Friuli Venezia Giulia and our site is located 113m above sea level which is lower than 1000m so we can take 25m/s $V_{b,0}$.

$$V_b = V_{b,0} \quad a_s \leq a_0$$

$$V_b = 25 \text{ m/s}$$

As data collected from website refer the same value of wind velocity.

PEAK VELOCITY PRESSURE:

Peak Velocity pressure can be found with this formula.

$$q_b = 1/2 \cdot \rho \cdot V_b^2$$

Where ρ is the density of air which is 1.25kg/m³ and V_b is the basic wind velocity.

$$q_b = 1/2 \cdot \rho \cdot V_b^2$$

$$q_b = 1/2 \cdot 1.25 \cdot 1000 \cdot 25^2$$

$$q_b = 390.625 \text{ N/m}^2$$

As website refers the same amount of velocity pressure 0.39 kN/m²

All the other parameters for wind loads are reported in this table.

WIND LOAD				
a_s [m]	113			
a_0 [m]	1000	→	v_b [m/s]	25
$V_{b,0}$ [m/s]	25			
k_a	0			
ρ_{aria} [N/m ³]	12.26	→	q_b [kN/m ²]	0.39
k_r	0.22			
c_t	1			
z_0 [m]	0.3			
z [m]	8.8	→	p [kN/m ²]	0.53

C _p	0.8	→	q_w [kN/m²]	0.23
C _d	1			
C _e	1.70			
α	0.52			

Table 10: Wind Load

Structural loadings acting on the elements are calculated as follow:

STRUCTURAL LOADS							
	G [kN/m ²]	Q [kN/m ²]	Ψ ₂	Q _{snow} [kN/m ²]	Ψ _{2,snow}	Q _{wind} [kN/m ²]	Ψ _{2,wind}
Slab	4.05	2	0.3	-	-	-	-
Roof	1.28	0.5	0	1.20	0	0.23	0
Stairs	3.1	4	0.6	-	-	-	-

Table 11: Structural Loads

2.2 SEISMIC ACTION:

Seismic action on structure is given by the uneven motion of the sediment soil, due to the propagation of seismic waves. Seismic motion excites the structure causing its dynamic response, which must be verified in terms of safety and performance parameters.

Behavior Factor: The behavior factor “q” is used to take into consideration the inelastic behavior of the building. This factor is computed based on NTC. In a more general way, the behavior factor is calculated using the formula below:

$$q = 2 \frac{\alpha_u}{\alpha_1}$$

- α_1 is the value by which the horizontal seismic design action is multiplied to first reach the flexural resistance in any member of the structure, while all other design actions remain constant.
- α_u is the value by which the horizontal seismic design action is multiplied, to form plastic hinges in several sections sufficient for reaching overall structural resistance, while all other design actions remain constant.

	EDIFICI NUOVI		EDIFICI ESISTENTI
	1 PIANO	2 o PIU' PIANI	QUALSIASI n° PIANI
REGOLARI IN ELEVAZIONE E IN Pianta	q = 2,8 $q_{0x} \cdot u^* \cdot 1 \cdot k_0 = 2 \times 1,4 \times 1$	q = 3,6 $q_{0x} \cdot u^* \cdot 1 \cdot k_0 = 2 \times 1,8 \times 1$	q = 3 $q_{0x} \cdot u^* \cdot 1 = 2 \times 1,5$
IRREGOLARI IN ELEVAZIONE E REGOLARI IN Pianta	-	q = 2,88 $q_{0x} \cdot u^* \cdot 1 \cdot k_0 = 2 \times 1,8 \times 0,8$	q = 2,25 $q_{0x} \cdot u^* \cdot 1 = 1,5 \times 1,5$
REGOLARI IN ELEVAZIONE E IRREGOLARI IN Pianta	q = 2,4 $q_{0x}(1+u^* \cdot 1)/2 \cdot k_0 = 2 \times (1+1,4)/2 \times 1$	q = 2,8 $q_{0x}(1+u^* \cdot 1)/2 \cdot k_0 = 2 \times (1+1,8)/2 \times 1$	q = 2,5 $q_{0x}(1+u^* \cdot 1)/2 = 2 \times (1+1,5)/2$
IRREGOLARI IN ELEVAZIONE E IN Pianta	-	q = 2,24 $q_{0x}(1+u^* \cdot 1)/2 \cdot k_0 = 2 \times (1+1,8)/2 \times 0,8$	q = 1,875 $q_{0x}(1+u^* \cdot 1)/2 = 1,5 \times (1+1,5)/2$

Tabella 6: sintesi dei valori da adottare per il fattore di struttura relativamente ad edifici in muratura (non armata) nuovi ed esistenti, in base alla regolarità in elevazione ed in pianta.

Figure 14- Recommended Values for Behavior Factor for Masonry

The considered building is analyzed being irregular in plan and regular in elevation. Value of the behavior factor based on the above-mentioned assumption is given as: q=3.

2.3 RESPONSE SPECTRUM:

Using Eurocode 8, the elastic response spectrum can be defined. This depends on the ground condition. The ability of the structural systems to respond to the seismic actions in their non-linear regime, allows for the design according to smaller actions than the ones provided by a theoretical linear elastic response. To avoid considering the inelastic structural analysis during design and to account for the ability of the structure to dissipate energy, due to the behavior of its ductile elements, an elastic analysis is performed but using the original elastic response spectrum divided by the behavior factor of the structure.

To compute the seismic forces acting on the structure, the response spectrum was calculated for the location of Udine Italy. The parameters of the considered building in this project are given as follows: -

- Class of use II
- Topographic category: T1
- Soil type: C
- Damping $\xi = 5\%$
- Return period $T_r = 475$ years

	Years	a_g	F_o	T_c^*
SLO	30	0.056	2.470	0.240
SLD	50	0.074	2.472	0.262
SLV	475	0.206	2.447	0.332
SLC	975	0.274	2.486	0.346

Table 12: Acceleration for various returning periods

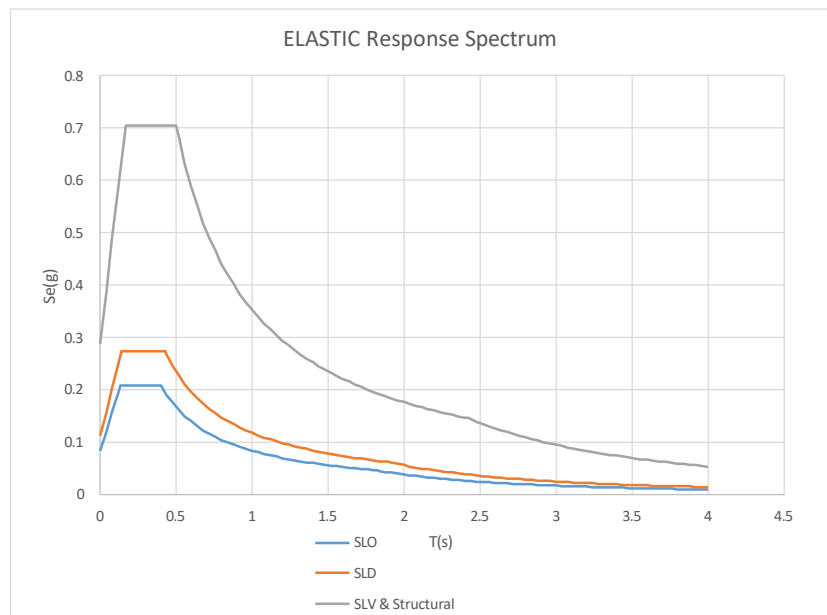


Table 13: Elastic response spectrum

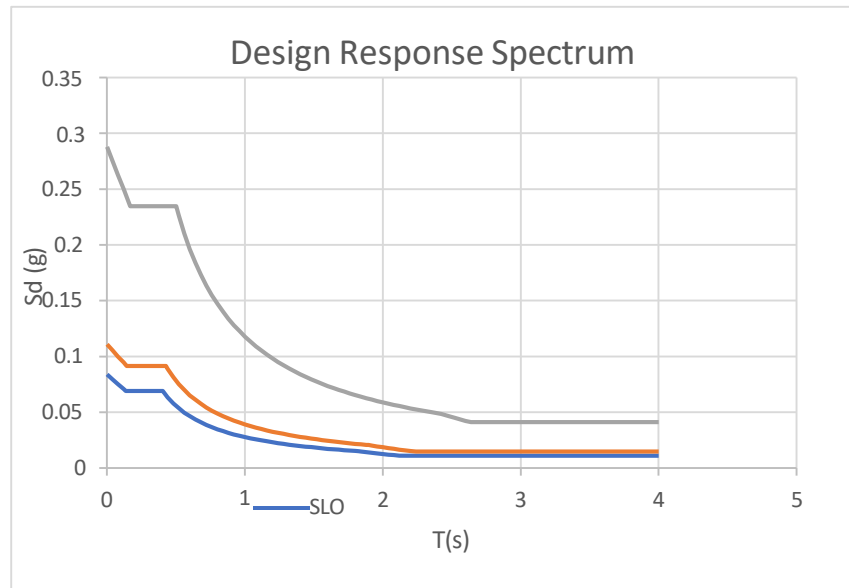


Table 14: Design response spectrum

2.4 PERIOD OF VIBRATION:

The value of the fundamental period of vibration can be approximated using the following formula:

$$T1 = C1 * H^{3/4}$$

- $C1$ is a coefficient that considers the type of construction. For masonry $C1 = 0.05$.
- The total height of the building, $H = 8.6\text{m}$

$$T1 = 0.251 \text{ s}$$

2.5 SEISMIC MASS:

The equivalent mass under the seismic action is calculated as:

$$\sum G_{kj} + \sum \psi_{E,i} Q_{ki}$$

Where:

- G_{kj} is the permanent loads,
- Q_{ki} is the variable loads and the coefficient
- $\psi_{E,i} = \varphi \cdot \psi_{2,i}$ depends on the function of the structure. For domestic building $\varphi=1.0$ for the roof and $\varphi = 0.5$ for independently occupied stories, while $\psi_{2,i} = 0.3$.

	Permanent			Variable		
	masonry	slabs	stairs	slabs	stairs	snow/wind
	G[kN]	G [kN]	G [kN]	Q [kN]	Q [kN]	Q [kN]
ground floor	3092.32	324	26.629	160	19.34	0
first floor	3092.32	118.2	26.629	184.72	0	166.2

Table 15: Permanent and variable loads

Factors that considered for total seismic masses are,

Element	slabs	roof	stairs	snow/wind
ψ_2	0.3	0	0.6	0

Table 16: Factors for seismic mass

Total Seismic mass is 692.67tons.

CHAPTER 3:

STRUCTURE ANALYSIS:

3.1 CENTRE OF GRAVITY:

The special importance for the behavior of the building should be given to the position of the center of gravity and the center of stiffness. The torsional effects being present in the building are as result of the eccentricity of the floor, considering the center of gravity and center of stiffness. The gravity center position is given based on the expressions below:

$$x_g = \frac{\sum G_i * x_i}{\sum G_i}$$

$$y_g = \frac{\sum G_i * y_i}{\sum G_i}$$

Where:

- x_i and y_i are the coordinates of the gravity center of each wall with respect to an arbitrarily chosen reference system
- G_i is the seismic weight of the element

As our building is rectangular in its section so our gravity center or mass center x_g and y_g will be 4, 5m respectively.

3.2 CENTRE OF RIGIDITY:

The position of the center of rigidity is computed in the following way:

$$x_s = \frac{\sum_k x_k * G_{M,k} * A_{yi,k}}{\sum_k * G_{M,k} * A_{yi,k}}$$

$$y_s = \frac{\sum_k y_k * G_{M,k} * A_{xi,k}}{\sum_k * G_{M,k} * A_{xi,k}}$$

Where:

- x_k and y_k are the coordinates of the gravity center of each wall with respect to an arbitrarily chosen reference system
- $G_{M,k}$ is shear modulus of the wall
- $A_{ji,k}$ is the area of the element i in the j direction

PEERS DIRECTION X				
Wall	A [m ²]	x _i [m]	y _i [m]	A _{xi,k} *y _k
WH1	3.71	4.00	9.74	36.14
WH2	1.08	6.75	6.24	6.71
WH3	2.42	5.39	4.24	10.26
WH4	3.18	4.00	0.24	0.76

Table 17: Peers in X direction

PEERS DIRECTION Y				
Wall	A [m ²]	x _i [m]	y _i [m]	A _{yi,k} *x _k
WV1	5.30	0.26	5	1.38
WV2	1.40	3.26	2.25	4.58
WV3	1.91	5.26	6.5	10.05
WV4	5.30	7.74	5	41.00

Table 18; Peers in Y direction

Ground floor, first floor			
Stiffness center		Mass center	
Y _s [m]	5.19	Y _G [m]	5.00
X _s [m]	4.10	X _G [m]	4.00

Table 19-Stiffness Center and Gravity Center

So, our eccentricities will be,

Eccentricity	
Ground, first floor	
e _y [m]	0.19
e _x [m]	0.10

CHAPTER 4

SAFETY EVALUATION:

Methods of evaluation (in general) they can be divided into two main groups: configuration-based checks and strength checks. The configuration-based checks serve as a tool to figure out unfavorable building characteristics, while the strength checks serve to assess the load carrying capacity of the structure. If the configuration is unfavorable, the building is not performing as well during an earthquake as a regular building. Deficiencies are found in irregular geometry, a weakness in the given story, concentration of mass, or in discontinuity in the lateral force resisting system. Strength, stiffness, geometry, and mass are found to involve irregularities, while horizontal distribution of lateral forces to the resisting frames or shear walls involve horizontal irregularities. As seen in the following figure, the checks include an investigation of the load path, weak story, soft story, geometry, and effective mass as well as torsion and pounding

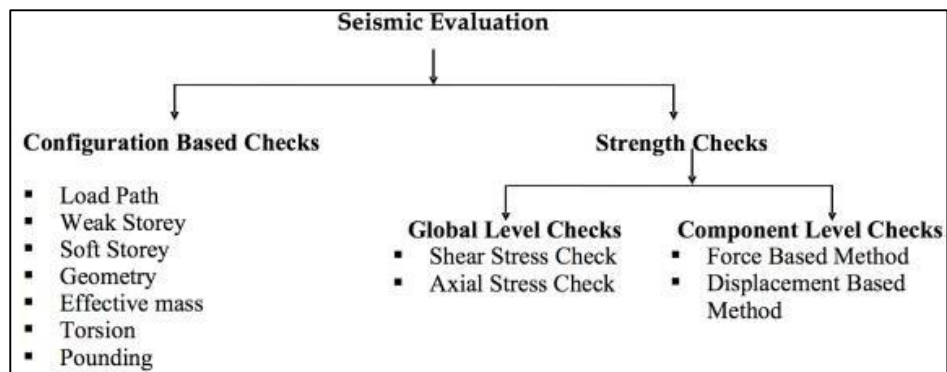


Figure 15- General Structure Evaluation Procedure

This project considers only level 1: Simplified global behavior analysis for seismic safety.

This project considers only first, the simplest, LV1 analysis. The concept of capacity vs. demand, which represents the base of this method, is illustrated in the picture below:

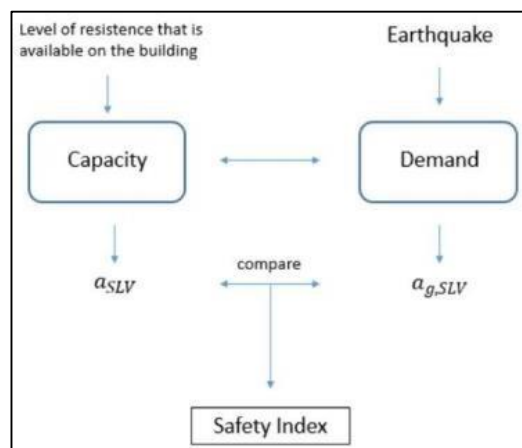


Figure 16- Schemes for Computing Safety Index

The capacity of the structure interpreted in terms of ground acceleration (α_{SLV}) is computed using the design response spectrum. The value of structural acceleration (S_e , SLV) corresponding to the limit collapse condition of the building:

$$S_{e,SLV} = \frac{q \cdot F_{SLV}}{e^* \cdot M}$$

Where:

- F_{SLV} is the shear resistance of the building
- e^* is the portion of mass participating in the first mode
- M is the total seismic mass

The hypothesis made for simplification assumes that the response of building is in its first natural mode which represents the distribution of displacement linearly to the height of the building.

The value of acceleration is computed using the following expressions:

$$\alpha_{SLV} \begin{cases} \frac{S_{e,SLV}(T_1)}{S \cdot F_0}, & \text{if } T_B < T_1 < T_C \\ \frac{S_{e,SLV}(T_1)}{S \cdot F_0} \cdot \frac{T_1}{T_C}, & \text{if } T_C < T_1 < T_D \end{cases}$$

In our case $T_B < T_1 < T_C$.

3.1 EVALUATION OF SHEAR RESISTANCE:

The Shear Resistance (F_{SLV}) in the two main directions of the building, x-x, and y-y, can be determined based on the formulations below,

$$F_{SLV,xi} = \frac{\mu_{x,i} \cdot \xi_i \cdot \zeta_i \cdot A_{x,i} \cdot \tau_{di}}{\beta_{x,i} \cdot \kappa_i}$$

$$F_{SLV,yi} = \frac{\mu_{y,i} \cdot \xi_i \cdot \zeta_i \cdot A_{y,i} \cdot \tau_{di}}{\beta_{y,i} \cdot \kappa_i}$$

Where:

- $\mu_{k,i}$ – is a coefficient that accounts for the stiffness and resistance homogeneity of the bearing wall i, in the direction k
- $\xi_i = 1$ – is coefficient related to the dominant type of collapse like shear collapse on floor i
- $\zeta_i = 1.0$ – is a coefficient related to the resistance of bearing the wall on the floor level i in this case we considered strong spandrel
- $A_{k,i}$ – is the area of the wall on the floor i, resisting to shear in the direction k
- $\tau_{d,i}$ – is the design value of shear resistance of the masonry of the wall, on the floor i
- $\beta_{k,i}$ – is a coefficient of irregularity in plan on the floor i, in the direction k
- κ_i – is the ratio between the shear force at the floor i and the total seismic force.

FACTOR κ_i :

This factor is the ratio between the shear force at the floor i and the total seismic force. So, in our case we have ground first and Roof Floor. Our Factors will be found by using this formula.

$$\kappa_i = \frac{\sum_i^N m_j \cdot \varphi_j}{\sum_1^N m_j \cdot \varphi_j},$$

Floor	j	κ_i
Ground	0	1
First	0.5	0.6666667
Roof	1	

Table 20- Factor κ_i **COEFFICIENT OF IRREGULARITY $\beta_{k,:}$:**

Coefficient which caused irregularity in plan is by eccentricities and found with these formulas in both direction for all floors.

$$\beta_{x,i} = 1 + \frac{e_{y,i} \cdot d_{y,i} \cdot A_{x,i}}{\sum_k (y_k - y_{R,i})^2 \cdot A_{xi,k}} \leq 1.25$$

$$\beta_{y,i} = 1 + \frac{e_{x,i} \cdot d_{x,i} \cdot A_{y,i}}{\sum_k (x_k - x_{R,i})^2 \cdot A_{yi,k}} \leq 1.25$$

Where:

- $e_{x,i}$ and $e_{y,i}$ – the eccentricities in the distance between the stiffness center and the center of mass in the direction X or Y respectively
- d_x , - the largest distance between the stiffness center and the external wall in Y direction. In analogous way for d_y , , the wall is considered is in the X direction.

Floor	Direction	e_k	d_k	β
F ₀ , F ₁	y	0.19	5.19	1.0721
F ₀ ,F ₁	x	0.10	4.1	1.04701844

Table 21- Irregularity coefficient

DESIGN VALUE OF SHEAR RESISTANCE OF MASONRY:

Design Value for shear resistance can be found by using these formulas.

$$\tau_{di} = \tau_{0d} \cdot \sqrt{1 + \frac{\sigma_{0i}}{1.5 \cdot \tau_{0d}}}$$

- τ_0 – the nominal shear resistance of masonry accounting for the confidence coefficient (FC)
- σ_0 – the average normal strength value on the resisting surface of the walls in the i floor.

For that we need stresses for all the elements.

STRESSES:

Masonry walls SW				Structural floor loads				Structural stair loads			
WALL	AM _{tot} [m ²]	G [kN]	σ Mpa	G [kNm ²]	A _f	N [kN]	σ Mpa	G [kN/m ²]	A _{stairs}	N [kN]	σ Mpa
0-1 floor	24.13	3092.33	0.13	4.05	80.0	324	0.01	3.10	4.84	15	0.0006
1-Roof	24.1249	3092.32924	0.1282	1.28	92.36	118.22	0.0049	3.10	0.00	0.00	0

Table 22: Stresses due to self-weight

Variable floor loads						Variable stair loads			
WALL	AM _{tot} [m ²]	G [kN]	σ Mpa	G [kN/m ²]	A _f	N [kN]	σ Mpa	G [kN/m ²]	A _{stairs}
0-1 floor	24.125	2.00	80.0	160.0	0.0066	4.00	4.84	19.34	0.0008
1-roof floor	24.125	2.00	92.4	184.7	0.0077	4.00	0.00	0.00	0.0000

Table 23: Variable floor and stair loads

Snow and wind loads			
Q [kN/m ²]	Area _{floor}	N [kN]	σ [MPa]
0.00	116.23	0.00	0.0000
1.43	116.23	166.20	0.0069

Table 24: Snow and wind loads

Total Stresses and loads are,

Walls	A _{Mtot} [m ²]	N _{tot} [kN]	σ [MPa]
0-1 floor	24.1249	7172.13	0.2973
1-roof floor	24.1249	3561.47	0.1476

Table 25: Total Stresses

After finding stresses by using the following formula we can find τ_{0d} .

$$\tau_{0d} = \frac{\tau_0}{F_c}$$

After finding all the parameters we can use above mentioned formula to find design shear resistance.

Design shear resistance τ_d is reported in this table.

Floor	σ ₀ [Mpa]	τ _{0d} [Mpa]	τ _d [Mpa]
Ground	0.297	0.007	0.038
First	0.148	0.007	0.027

Table 26- Design Shear Resistance

- $\mu_{k,i}$ is a coefficient that accounts for the stiffness and resistance homogeneity of the bearing wall i, in the direction

$$\mu_{k,i} = 1 - 0.2 \cdot \sqrt{\frac{N_{m,xi} \cdot \sum_{j=1}^{N_{m,xi}} A_{ki,j}^2}{A_{x,i}^2}} - 1 \geq 0.8$$

Where:

- $N_{m,ki}$ is the number of walls in the k direction on the floor i.
- $A_{ki,j}$ is the area of the wall j in the direction k, on the floor i.

Direction y				
Floor	ΣA_y^2	A_{yi}^2	N_m	μ
F ₀ , F ₁	61.80	193.60	4	0.89

Table 27: Stiffness and resistance homogeneity coefficient in Y direction

Direction x			
ΣA_x^2	A_{xi}^2	N_m	μ
30.89	107.85	4	0.92

Table 28: Stiffness and resistance homogeneity coefficient in X direction

The Shear Resistance (F_{slv}) in the two main directions of the building, x-x, and y-y are found by the formula as all the coefficient for that are already found. F_{slv} are reported below,

	Floor	μ	ξ	ζ	A	τ_d	β	k_i	F_{SLV} [KN]
X	Ground	0.92	1	1	10385000	0.0377	1.05	1.00	345.0
	First	0.92	1	1	10385000	0.03	1.05	0.67	370.8
	Floor	μ	ξ	ζ	A	τ_d	β	k_i	F_{SLV} [KN]
Y	Ground	0.89	1	1	13914000	0.04	1.07	1.00	437.3
	First	0.89	1	1	13914000	0.03	1.07	0.67	470.1

Table 29: Shear resistance in X and Y direction

$$S_{e,SLV} = \frac{q \cdot F_{SLV}}{e^* \cdot M} \quad e^* = \frac{(\sum_i^N m_j \cdot \varphi_j)^2}{M \cdot \sum_1^N \varphi_j^2}$$

F_{SLV} [kN]	344.997
q	3
e^*	0.8987
M [ton]	692.67
S_{eSLV} [m/s ²]	1.6627
S_{eSLV} [g]	0.1695

Table 30: Spectral acceleration

3.2 RESPONSE SPECTRUM OF THE STRUCTURE:

Once the global shear resistance of the building (F_{SLV}), which corresponds to the minimum shear resistance between the two main directions, the acceleration corresponding to the elastic response spectrum at ultimate limit state SLV (“Stati Limite di Salvaguardia della Vita”) ($S_{e,LV}$) is determined.

$$a_{slv} = \frac{S_{e,slv}(T)}{S * F_0}$$

Therefore, the return period (TR) corresponding to the acceleration can be obtained approximately through linear interpolation:

Interpolation Rule:

$$\log p = \log p_1 + \log \frac{p_2}{p_1} * \frac{\log (T_r/T_{r1})}{\log (T_{r2}/T_{r1})}$$

Where:

- p can be any parameter of the following seismic parameters: design ground acceleration (a_g), maximum value of the amplification factor of the spectrum (F_0) or corner period at the upper limit of the constant acceleration region (T^*C).

The exact return period corresponding to the ground acceleration can be obtained by performing an iterative procedure. Performing this iteration leads to the return period of 17 years.

SLD		SLV	
TR [years]	50	TR [years]	475
a_g [g]	0.074	a_g [g]	0.206
F_0	2.472	F_0	2.447
T_c^* [s]	0.262	T_c^* [s]	0.332
S_s	1.5	S_s	1.5
C_c	1.634	C_c	1.511
ST	1	ST	1
q	1	q	1
S	1.5	S	1.5
η	1	η	1
TB [s]	0.1427	TB [s]	0.1672
TC [s]	0.428	TC [s]	0.50159
TD [s]	1.896	TD [s]	2.424

Table 31: Limits of period in response spectrum

3.1 EQUIVALENT RESPONSE SPECTRUM:

Equivalent response spectrum and limit values of time periods are given in the following table.

EQUIVALENT SPECTRUM	
TR [years]	17
a_g [g]	0.045
F_0	2.484
T_{c^*} [s]	0.234
S_s	1.5
C_c	1.471
S_T	1
q	1
S	1.5
η	1
T_B [s]	0.115
T_c [s]	0.344
T_D [s]	1.781

Table 32: Equivalent spectrum

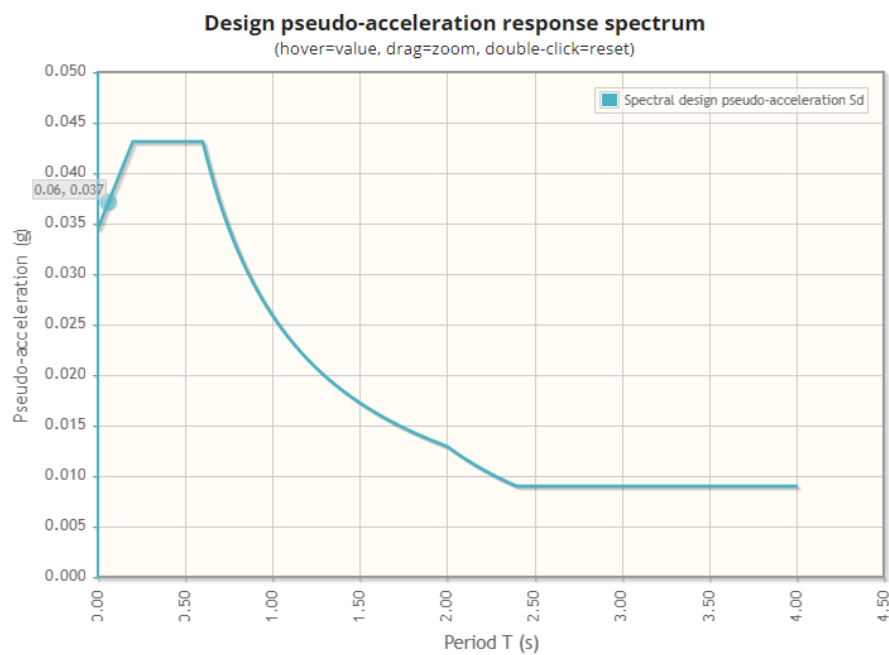


Figure 17- Design Spectrum for ULS

3.1 SAFETY FACTOR:

Finally, in order to evaluate the vulnerability of the structure, the acceleration factor ($a_{s,SLV}$) and seismic safety index ($I_{s,SLV}$) related to the limit state SLV provide a non dimensional number in order to evaluate the existing structure.

CAPACITY	
$a_{s,SLV}$ [g]	0.045
T1 [s]	0.251
$S_{e,SLV}$ [g]	0.169

DEMAND	
a_g SLV [g]	0.206
TR [annual]	475

$$a_{s,SLV} = \frac{a_{s,SLV}}{a_{g,SLV}} = \frac{0.045g}{0.206g} = 0.218 < 1$$

$$I_{s,SLV} = \frac{T_{s,SLV}}{T_{r,SLV}} = \frac{17}{475} = 0.04 < 1$$

CONCLUSIONS

As mentioned above the design approach for the evaluation of this existing building is used the LV1 approach. This type of analysis is performed following some assumptions which are practical for preliminary evaluation of the existing buildings.

Considering lack of provided information there are several aspects and reference values which must be taken into consideration including type and mechanical properties of masonry (elastic modulus, shear modulus). These values are taken as the minimum ones for the safety reasons. By providing experimental testing for the mechanical properties of the material leads to the more reliable analysis and accurate results.

Values calculated for the acceleration factor ($a_{s,SLV}$) and seismic safety index ($I_{s,SLV}$) are lower than a unit value which leads to the conclusion that the structure is not complying with the current provisions. This might be reasonable because the analyzed structure might have been designed with old codes in which the concepts of return period and probability of failure were not well developed.

. The LV2 and LV3 are more complex approaches used in cases when required to have more accurate results by defining the properties of the material in a more refined way.

References:

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